A Computational Framework to Uncover Gray Areas in Tax Legislation

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Abstract. It is often said that every law has its loophole, meaning that legal rules frequently contain technical flaws, ambiguities, or unspecified details that can be legally exploited to bypass the intent of the law. In the context of taxation, such gray areas contribute to regressivity, unfairness, and substantial revenue losses, costing governments and companies billions of dollars worldwide. Reducing tax disputes increasingly requires more efficient and cost-effective strategies than traditional ex post resolution. In this context, natural language processing models offer a scalable and adaptable opportunity to anticipate gray areas directly from legal texts. However, systematic approaches that leverage these tools to identify potential sources of tax norm abuse at scale remain limited. This paper asks whether language-based computational methods can contribute to the systematic identification of gray areas in tax legislation. To address this question, we propose a methodology to define, identify, and annotate legal provisions according to their potential gray-zone risk. We apply this methodology to Colombian Free Trade Zone (FTZ) legislation and construct, to our knowledge, the first systematic dataset of tax norms annotated for gray-area-related non-compliance risk. Using this dataset, we implement classification tasks based on TF–IDF representations, compare its performance with transformer-based models, and relate it to tax law complexity. The classification achieves around 80% F_1 scores at its best specification.

Keywords: Legal NLP · Computational Legal Analysis · Tax Law Analysis · Legal Text Classification

1 Introduction

A common saying states that every law has its loophole. Colombian Free Trade Zone legislation offers a clear illustration of how loopholes do not disappear with reform, but rather continue to manifest in different contexts over time. In 1958, Free Trade Zones were introduced as a policy instrument to stimulate investment and employment through the production of manufactured goods in Colombia. Law 105 of 1958 granted tax benefits to firms established in delimited coastal areas, under the expectation that these firms would engage in productive activity [1]. The design of the laws, however, allowed domestic firms to locate in these zones, access preferential treatment, and import goods without carrying out any real production. Subsequent reforms attempted to close this gap. By the mid-1980s, legislation distinguished between commercial and industrial zones, granting tax benefits only to firms that "develop the industrialization process of a product".[1] Because the notion of industrialization remained undefined, firms carried out purely formal transformations, such as labeling or repackaging, to qualify for the benefit. More recent reforms followed the same pattern. Given that 75 percent of firms operating in Free Trade Zones produced for the domestic market [10], Law 2277 of 2022 replaced production requirements with the obligation to submit an export plan reviewed by a public official [2]—shifting legal compliance to individual assessment, which renders access to the benefit dependent on discretionary power. Each reform closed a visible loophole and opened a new one, rooted in legal ambiguity.

Ambiguity, incompleteness, discretionary powers, and differential treatments in law generate gaps where legal interpretation determines tax outcomes. This results in tax controversies that impose high costs on taxpayers and administrations [11]: In 2022, firms involved in transfer pricing disputes reported average expenditures of USD 56.3 million in assessments, USD 24.7 million in penalties and interest, and USD 21.3 million in legal fees [12]. In 2025, 825 surveyed businesses reported at least one recent tax dispute, and over 70 percent indicated an increase in tax audits in recent years as governments seek to close loopholes and curb avoidance [27]. Through these gaps, firms pay less tax, public revenue declines, and fiscal capacity erodes. Such disputes reflect a reactive enforcement paradigm that intervenes after exploitation occurs, rather than preventing it at the drafting stage.

Despite its relevance, the problem remains weakly systematized in the literature. Governments and tax authorities address gray areas on a case-by-case basis, often through audits, litigation, or ad hoc administrative guidance. Computational approaches that attempt to anticipate gray areas directly from statutory language remain in an early stage. While recent studies explore how language models interact with tax law, there is still no systematic framework that links linguistic features of legal texts to the risk of norm abuse at scale.

This paper addresses the following research questions: How can gray areas in tax legislation be defined in a way that is both legally meaningful and computationally tractable? Can linguistic features of statutory texts signal provisions that are prone to exploitative interpretation? To what extent can language-based models recover expert assessments of gray-area risk, and where do they fail? How stable are the predictions when the input is modified through synonym substitution and phrase reordering? What is the relation between tax law complexity and the gray area risk detection?

The paper contributes to the literature in three ways. First, it proposes an operational framework to identify gray-area indicators grounded in legal theory and expert practice. Second, it constructs the first systematic dataset of Colombian tax provisions annotated for gray-area-related risk, using Free Trade Zone legislation as a case study. Third, it evaluates the ability of standard text representations and legal language models to detect these indicators, and relates their performance to legal text complexity. The remainder of this paper is organized as follows: Section 2 provides the theoretical background on legal uncertainty. Section 3 reviews related work. Section 4 details the proposed framework, dataset construction, and empirical results. Section 5 discusses the implications and limitations of the study, and Section 6 presents the conclusions.

2 Background

Legal uncertainty is a structural and unavoidable feature of tax law rather than a drafting defect. While the principle of certainty of taxation functions as a normative ideal, aiming at clarity, predictability, and stability, it cannot be fully realized in practice. As argued by Demin [11], legal uncertainty operates along a continuum and may arise both negatively, through omissions or inconsistencies, and positively, as a regulatory technique that enables flexibility and case-specific judgment in complex economic environments. Core sources of legal uncertainty in tax law include:

Lack of accuracy and clarity: Uncertainty arises from imprecision in legal drafting, limited understandability and accessibility, and incomplete regulation, including gaps in statutory provisions. It also reflects tensions between abstract norms and concrete applications; e.g., *undefined taxable events*, *partial regulatory coverage*.

Instability and inconsistency: Tax law is characterized by chronic instability, inconsistency, and fragmentation; e.g., *multiple amendments* may affect the same tax provision.

Vague or evaluative terms and concepts: *Open-textured* legal concepts lack fixed meaning and require interpretation by taxpayers, administrators, and courts. Their content is shaped through individual decoding and subsequent legal practice; e.g., terms such as *reasonable*, *ordinary* and *necessary expenses*

Open-ended statutory lists: Enumerations that are explicitly non-exhaustive function similarly to vague concepts, allowing authorities and courts to extend their scope with additional elements; e.g., expressions such as *other economically equivalent transactions* and *any similar arrangement.*

Discretion: Discretion arises when the law authorizes officials to depart from general rules based on subjective assessments of specific circumstances; e.g., *approval subject to* the authority's assessment.

In this context, we define **tax avoidance strategies** as the deliberate exploitation of these structural uncertainties—such as ambiguities, incomplete regulations, or discretionary powers—to minimize tax liabilities while remaining strictly within the formal boundaries of the law, distinguishing it from illegal tax evasion.

3 Related Work

Detecting exploitable ambiguities in legal texts requires understanding how language creates interpretive flexibility that enables norm abuse. We organize prior research into three areas: natural language processing approaches to ambiguity, computational methods targeting tax uncertainty, and alternative techniques using genetic algorithms.

3.1 NLP and Ambiguity Detection

The challenge of computationally disambiguating meaning in text has long been recognized as difficult. Navigli [22] shows that fine-grained *disambiguation*, understood as "the ability to identify the meaning of words in context in a computational manner", achieves only 70% accuracy, establishing fundamental limits for semantic tasks. Bhattacharya et al. [3] probe BERT and GPT-2 representations and find that *ambiguity* is poorly encoded, with performance barely exceeding baselines. They further show that middle layers outperform upper layers, suggesting information loss at higher abstraction levels. Wildenburg et al. [26] deals with *underspecification* and demonstrate that models recognize underspecification at rates above chance, yet fail to identify cases in which underspecified sentences allow open-ended continuations (e.g., "among others").

In legal and technical domains, the picture becomes more complex. Guitton et al. [17] develop human annotation protocols for *open-texture* in EU legal texts, achieving a Cohen's Kappa of $\kappa = 0.70$ after reconciliation, up from an initial baseline of 0.24. This reveals both the difficulty and tractability of annotating abstract legal concepts, even for experts. In a posterior work, they use gpt-3.5-turbo and LLaMa-2-70b-chat on a corpus annotated and obtain 0.84 and 0.67 f1 scores [16]. Ferrari and Esuli [14] use domain-specific word2vec models to identify ambiguous terms across Wikipedia corpora, achieving Kendall's τ up to 0.88 for two-domain comparisons, though non-expert annotators miss domain-specific nuances. Ezzini et al. [13] achieve approximately 80% precision detecting ambiguity in requirements engineering, with domain-specific corpora improving detection by approximately 33%. Kim et al. [20] develop task-specific embeddings for smart speaker command ambiguity, finding that they outperform off-the-shelf embeddings. Taken together, these studies show that domain-specific representations consistently outperform generic ones, that annotation of abstract properties remains challenging, and that distinguishing whether models capture genuine semantic properties or merely surface lexical patterns remains an open question.

3.2 Computational Approaches to Tax Uncertainty

Recent studies show that language models exhibit non-trivial capabilities in analyzing legal texts, while consistently diverging from expert judgment. This divergence suggests that such models are better suited to augment, rather than replace, expert-driven approaches. Blair-Stanek et al. [4] show that OpenAI's `o1` model can generate novel tax strategies. However, their subsequent evaluation [5] reveals that Spearman's ρ for whole-strategy grading ranges only between 0.30 and 0.48 when compared to expert grades, far below the 0.89 achieved for simpler tasks. Guitton et al. [16] find that `gpt-3.5-turbo` detects *open-texture* in GDPR text with $F_1 = 0.84$, yet only 42% of the terms flagged by humans appear in model output, and inter-annotator agreement among LLMs remains low (κ = 0.01–0.36). Fratrič et al. [15] combine large language models with Prolog-based planning to expose loopholes in multinational structures, though the formalization of legal rules by LLMs remains insufficient without human fine-tuning.

3.3 Alternative Computational Methods: Genetic Algorithms

Hemberg et al. [18] analyze a loophole created by a modification to Section 754 of the U.S. Tax Code and show that "each time the IRS changes the tax code, tax evaders respond by exploiting new ambiguities". They employ co-evolutionary genetic algorithms to model adversarial taxpayer–auditor dynamics, in which tax schemes and audit rules evolve simultaneously. Extending this anticipatory approach to Special Economic Zones, recent work [21] applies Genetic Algorithms to simulated supply chains, demonstrating how structural rate differentials enable complex, compliant-appearing profit-shifting strategies.

3.4 Synthesis and Identified Gaps

In summary, while prior literature has extensively explored the theoretical dimensions of legal uncertainty, treating ambiguity, open-texture, and tax uncertainty as broad, abstract phenomena, a significant gap remains in translating these concepts into verifiable computational tasks. Existing NLP approaches often struggle to capture domain-specific legal nuances, and recent LLM applications in tax law frequently lack grounded, expert-driven frameworks to formally evaluate structural loopholes. This paper bridges this gap by crystallizing these broad, nebulous theoretical concepts into a concrete, systematic, and empirically observable set of gray-area indicators. Our framework advances beyond theoretical discussions of ambiguity to provide a computationally tractable tool for the automated detection of tax norm abuse by operationalizing legal uncertainty into discrete categories that can be consistently identified by both human annotators and computational models.

4 Methodology and Results

4.1 Framework for Gray-Area Risk Classification and Dataset Creation

As mentioned in the background, legal uncertainty has been defined in different ways across fields. We propose an approach that operationalizes specific manifestations of legal uncertainty in ways that are computationally tractable and empirically observable. The translation method is a qualitative operationalization process combining discussions with tax law experts, reconstruction of historical avoidance schemes, and review of the legal uncertainty literature. From these inputs, we mapped abstract theoretical concepts into a set of discrete features that are tangible, precise, and amenable to explicit decision rules, which were then consolidated into the indicators reported in Table 1 and further detailed in the technical appendix (Section 7). Although the presence of an indicator does not constitute avoidance per se, it allows us to systematically identify risk-prone features embedded in tax legislation.

Building on the formal framework introduced above, we collapse the structural components of legal gray areas into a finite set of operational indicators. A more detailed explanation of how the indicator gives rise to the risk of tax avoidance, as well as examples of each category can be found in the technical appendix.

The corpus comprises all legal norms governing Colombia's Free Trade Zone (FTZ) regime from 1958 to 2005. This legislative domain was selected because it has been repeatedly identified as a focal point for tax avoidance strategies and exhibits frequent regulatory reforms that often close existing loopholes while simultaneously creating new ones [10]. Each article was manually annotated by a subject matter expert for the presence of gray area indicators following predefined annotation guidelines and decision rules. `Relevance`, `completeness`, `discretion`, and `differential regime` were treated as predicates (true for presence, false for absence). `Interpretation` was annotated using an ordinal scale reflecting increasing degrees of interpretive openness: low(1), moderate(2), and high(3). More details on the annotation protocol and the full database can be consulted through the technical appendix.

The expert-annotated labels for each indicator exhibit a clear but modest class imbalance, with a systematic overrepresentation of the label `false`. This imbalance motivates the use of macro-averaged evaluation metrics, as they weight all classes equally and prevent overall performance measures from being dominated by the majority class. Figure 1 reports the presence of indicators in training dataset articles.

4.2 Embedding-Based Analysis

To evaluate whether gray-area triggers form distinct patterns beyond surface-level text, we map the provisions into a semantic space—a mathematical representation where texts with similar legal meanings are clustered together. We generate these dense vector representations using MEL [24], a benchmark transformer model

Indicator	Answers	Question
Completeness	Yes/No	Does the article contain a fragment that specifies legal or tax treatment for some cases while omitting other relevant situations?
Differential Regime	Yes/No	Does the article contain a fragment that assigns different tax treatments within the same system based on specific attributes or classifications?
Discretion	Yes/No	Does the article contain a fragment where the application or enforcement of tax rules depends on administrative judgment rather than explicit legal criteria?
Interpretation	Low/Medium/High	Does the article contain a fragment whose wording admits multiple plausible interpretations leading to distinct legal or tax outcomes?
Relevance	Yes/No	Does the article contain a fragment that directly affects tax liabilities or economic operations relevant for tax calculation?

Table 1: Annotation indicators and labeling scheme

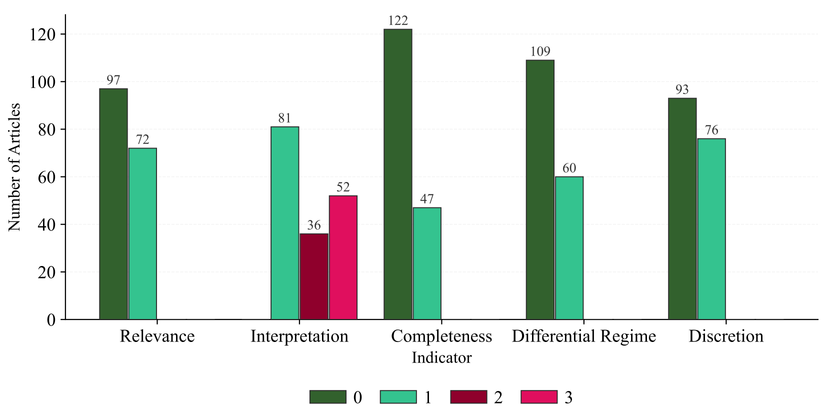


Fig. 1: Presence of indicators in training dataset articles

trained on Spanish legal texts to capture semantic nuances. Unlike bag-of-words approaches, these embeddings are designed to capture deep semantic properties. We then apply UMAP, a dimensionality reduction technique, to project these high-dimensional vectors into a 2D plane for visual inspection. Figure 2 presents these UMAP projections. `Interpretation` and `discretion` show no clustering, suggesting heterogeneous definitions or expert criteria not encoded in surface-level semantics. By contrast, `relevance`, `completeness`, and `differential regime` exhibit partial structure, indicating closer alignment between annotations and latent semantic space.

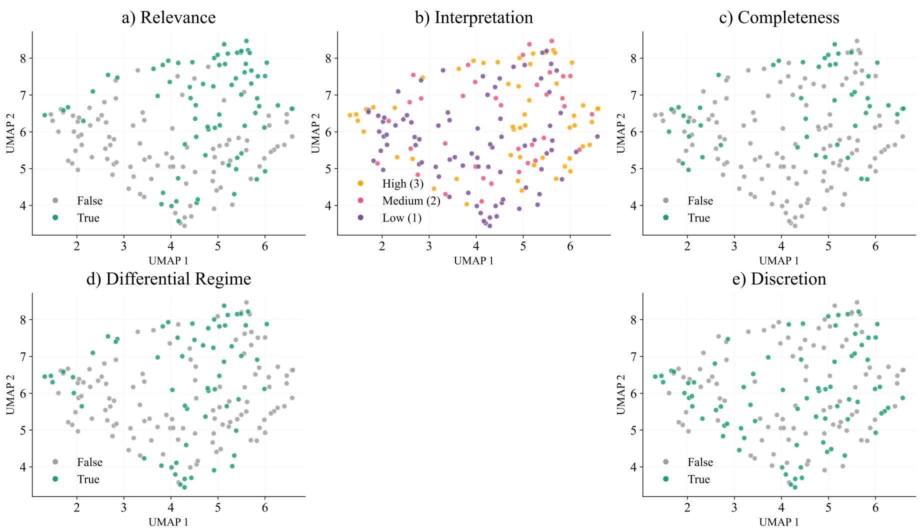
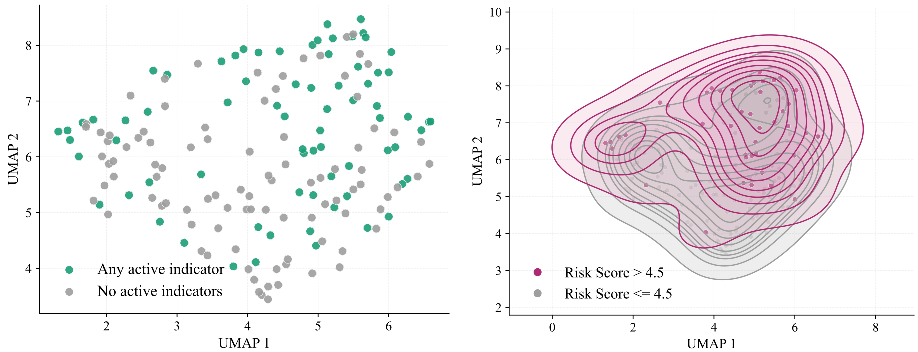


Fig. 2: Scatter plots of embeddings using MEL and UMAP.

We construct an aggregate risk score as a descriptive count of how many gray-area indicators are active in each article. Formally, for article i,

$$R_i = \sum_{k=1}^{K} \mathbb{1}_{\{p_{ik}=\text{true}\}} \tag{1}$$

where $K = 5$ is the total number of indicators, p_{ik} denotes the annotation assigned to indicator k in article i, and $\mathbf{1}\{\cdot\}$ is the indicator function. Thus, R_i ranges from 0 to 5 and can be interpreted simply as the number of active gray-area indicators in article i. Figure 3 compares articles with at least one active indicator and those with higher aggregated risk scores against low-risk articles in the embedding space. Both scatter and KDE plots show substantial overlap across groups, with no clear clustering or density separation for high-risk provisions.



(a) Scatter plot (b) UMAP KDE (MEL) - Risk Score > 4.5

Fig. 3: Global risk assessment of high-risk versus low-risk articles in the embedding space using MEL and UMAP.

4.3 Classification Task

We evaluate whether language models can learn expert-informed assessments of gray-area risk. To do so, we contrast models that capture surface lexical patterns (i.e., exact word frequencies and syntactic markers) with those designed to capture deep semantic properties (i.e., contextual legal meaning). We compare multiple text representation strategies, including TF-IDF as our lexical baseline, Legal-BERT on its light Spanish variant [7], and MEL [24]. Logistic regression is used as the classification model across all configurations, which provides an interpretable and widely adopted baseline in legal NLP applications [8,9]. The train/test proportions were 80 and 20% (135 and 34 articles respectively) Table 2 shows similar performance across models, with F1 scores around 0.7 for most binary indicators. TF–IDF remains competitive and achieves the best performance for `differential regime`, indicating that this indicator is largely captured by surface-level textual features. Domain-specific transformers provide limited gains: Legal-BERT improves `completeness` and `discretion`, while MEL performs best on `interpretation` and `relevance`. Overall, improvements from domain adaptation and Spanish-language training are modest and concentrated in semantically driven indicators, while most predictive signal appears to be lexical.

Table 2: Model performance comparison by indicator

	TF–IDF			Legal-BERT			MEL		
Indicator	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
`Completeness`	0.814	0.702	0.730	0.775	0.809	**0.788**	0.738	0.753	0.744
`Diff. Regime`	0.868	0.890	**0.875**	0.709	0.701	0.704	0.807	0.807	0.807
`Discretion`	0.766	0.754	0.757	0.793	0.788	**0.790**	0.674	0.675	0.674
`Interpretation`	0.262	0.343	0.272	0.331	0.362	0.333	0.455	0.457	**0.447**
`Relevance`	0.757	0.757	0.757	0.749	0.700	0.704	0.800	0.771	**0.779**

`Note:` Identical data were used for all three models ($n = 169$ in each of the 5 categories). All metrics are macro-averaged.

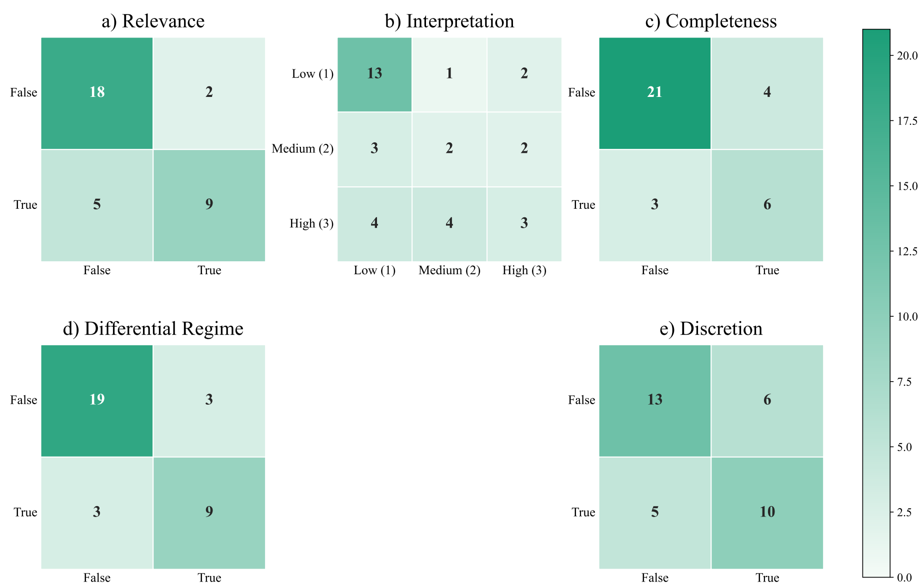


Fig. 4: Confusion matrices for MEL across the five indicators.

4.4 Robustness Analysis

Following [23] and [25], we assess robustness through controlled perturbations designed to separate sensitivity to surface form from sensitivity to core regulatory meaning. We generate two variants of each text. First, phrase reordering shuffles sentence-level units to alter syntactic organization while keeping most lexical material unchanged, testing whether predictions depend on structural arrangement. Second, lexical substitution replaces a fixed proportion of non-stopword tokens with Spanish WordNet synonyms, modifying surface lexical patterns while aiming to preserve the core meaning of the provision. In our annotation setting, these perturbations are treated as approximately label-preserving because indicators such as discretion or completeness depend on the regulatory mandate expressed by the provision rather than on exact clause order or synonymous lexical choices. Perturbed texts are embedded using MEL, and predictions are obtained from the trained model. Across the 170 perturbed test instances (test size 34 for each of the 5 categories), predictions remain unchanged in all but four cases: two for `discretion` and two for `relevance`, all induced by phrase reordering rather than lexical substitution. This indicates that model decisions are largely invariant to lexical variation and only marginally sensitive to syntactic reorganization, suggesting limited reliance on surface-level cues and weak sensitivity to higher-order structural changes.

4.5 Legal Complexity and Predictive Performance

To examine whether gray-area risk is associated with the textual and structural burden of legal provisions, we construct an exploratory article-level complexity proxy. Building on broader discussions of tax complexity [19], our goal is not to measure readability in the narrow psycholinguistic sense, but to approximate three observable dimensions of legal complexity: textual length, syntactic packing, and intertextual dependency.

Let i denote a legal article. We define W_i as the total number of words, S_i as the number of sentences, $L_i = W_i/S_i$ as the average sentence length, and R_i as the number of explicit normative references. These components are intended to capture, respectively, the overall volume of textual material, the density with which conditions and qualifications are packed into sentences, and the extent to which interpretation depends on cross-references to other legal provisions.

Each component is normalized using min-max scaling:

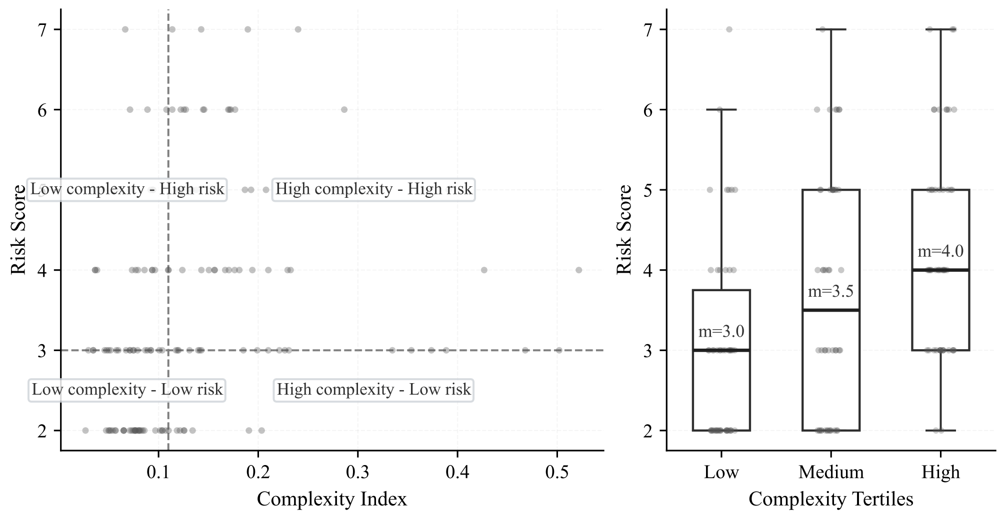
$$\tilde{X}_i = \frac{X_i - \min(X)}{\max(X) - \min(X)}, \quad X \in \{W, L, R\}. \tag{2}$$

We then define the textual complexity index as the unweighted average:

$$\text{Complexity}_i = \frac{1}{3}\left(\tilde{W}_i + \tilde{L}_i + \tilde{R}_i\right). \tag{3}$$

In the absence of an established theoretical or empirical basis for assigning different weights to these dimensions, we use equal weights to avoid imposing arbitrary assumptions about their relative contribution to article-level legal uncertainty. While future work could explore component-level ablations to isolate their individual contributions, we interpret this linear combination as an exploratory proxy for the overall textual and interpretive burden imposed by the provision rather than a fully validated measure.

Figure 5a shows that articles with lower complexity levels tend to be associated with lower risk scores, while articles with higher complexity are more frequently linked to higher risk scores. The plot partitions the space using the midpoints of the complexity index and the risk score, yielding four quadrants that facilitate a descriptive comparison between low- and high-complexity provisions and their corresponding risk levels. Figure 5b presents the distribution of risk scores across different levels of legal complexity. The results show that higher complexity levels are associated with higher median risk scores. In addition, the dispersion of the risk score increases with complexity, indicating greater variability in uncertainty among more complex legal provisions.



(a) Complexity index and risk score **(b)** Risk by complexity tertile

Fig. 5: Association between legal complexity and aggregate risk score.

Figure 6 shows the distribution of the complexity index across indicators. Notably, the indicators with weaker classification performance, namely `interpretation` and `discretion`, exhibit a wider dispersion in complexity values. In contrast, indicators with stronger classification results display comparatively lower variability in complexity, particularly for `relevance`, and to a lesser extent for `differential regime` and `completeness`.

5 Discussion

As noted above, this study examines gray-area triggers linked to expert-identified sources of tax norm abuse as a first step toward anticipating exploitative uses directly from statutory language. The analysis is intentionally restricted to uncertainty-based textual features and does not incorporate other mechanisms highlighted in prior work, such as interactions across legal provisions or enforcement dynamics [6].

The empirical results show clear differences across gray-area indicators. The strongest performance is observed for Differential Regime, Relevance, and Completeness. These indicators exhibit structured organization in the embedding space and consistent predictive behavior, indicating that they align with stable linguistic regularities present in statutory text. In contrast, Interpretation and

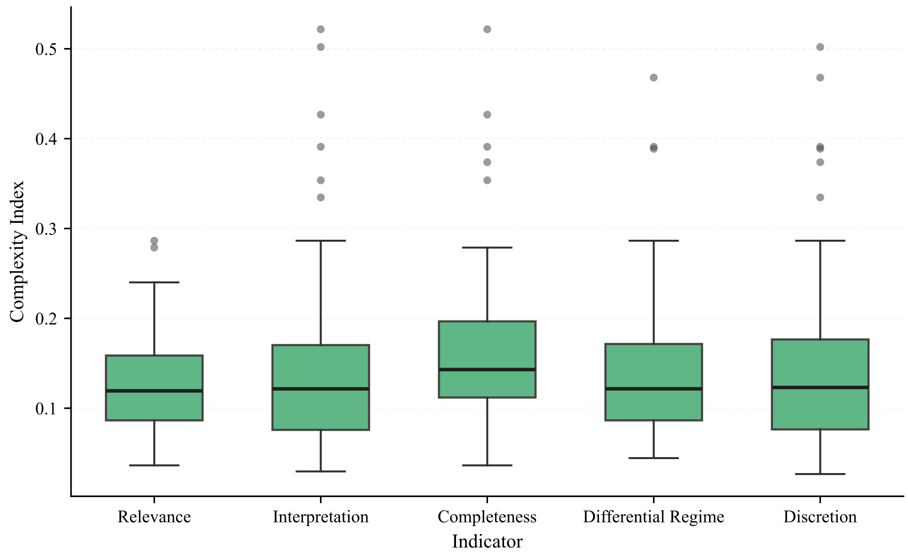


Fig. 6: Complexity index distribution by indicator

Discretion display weak predictive performance and lack discernible structure in vector representations. This pattern suggests that expert assessments for these latter indicators do not rely on textual properties encoded in the statutory language itself.

The absence of structure for Interpretation and Discretion indicates that the exploitability of these gray areas depends on contextual, institutional, or strategic factors external to the written norm. This finding supports prior literature arguing that uncertainty-based textual approaches should be complemented with methods that account for cross-provision interactions and enforcement signals in order to capture these dimensions of norm abuse.

At the aggregate level, the analysis also reveals limitations of the risk score construction. Since the overall risk score is defined as the simple sum of indicators-level indicators, the lack of separation in the embedding space suggests that aggregated risk is not reflected in global linguistic similarity. This result does not undermine the validity of expert annotations, but rather indicates that the criteria underlying risk assessment are not jointly encoded in the textual representations used in this study. We therefore conclude that a linear aggregation of indicators is not well suited to characterize overall gray-area risk, likely because the indicators do not exhibit linear independence and do not define a coherent group of provisions when combined.

6 Conclusion

A folk saying holds that every rule has its loophole. In tax law, the exploitation of such loopholes generates substantial economic costs for both taxpayers and tax administrations. Addressing this challenge requires scalable approaches capable of anticipating the emergence of gray areas before they are systematically exploited.

This paper contributes to that objective in three ways. First, it proposes an operational framework to identify gray-area indicators grounded in legal theory and expert practice, translating specific manifestations of legal uncertainty into indicators that are computationally tractable and empirically observable. Second, building on this framework, it introduces the first systematic dataset of Colombian tax provisions annotated for gray-area risk, using Free Trade Zone legislation as a case study. The annotation reveals that the indicators Relevance, Completeness, and Differential Regime capture coherent and related groups of provisions, whereas Interpretation and Discretion do not exhibit such structure.

Third, using this corpus, the paper evaluates the ability of standard text representations and legal language models to detect gray-area indicators. The results show that transformer-based architectures, including MEL, do not substantially outperform bag-of-words representations, suggesting that the annotations are primarily grounded in lexical rather than semantic features. In the best-performing specifications, classification achieved F1 scores of 0.88 for Differential Regime, 0.78 for Completeness, 0.79 for Discretion, 0.77 for Relevance, and 0.44 for Interpretation. Additionally, legal text complexity is positively associated with the presence of gray-area indicators and negatively associated with classification performance. This finding underscores the need to treat textual complexity as an explicit dimension in future computational approaches, in order to avoid biased or confounded detection outcomes.

The findings show that models operating on legal text can anticipate gray-area triggers grounded in explicit linguistic structure, including differential treatment, scope delimitation, and normative completeness. In these cases, the statutory language itself provides sufficient signal for systematic identification, which provides a basis for future computational work on the anticipation and detection of tax avoidance strategies.

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7 Appendix

Due to space constraints, the complete formalization underlying the framework is provided in an external technical document, available at: technical-appendix.

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